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RESEARCH ARTICLE



# How Philosophical Beliefs about Science Affect Science Education in Academic Engineering Programs: the Context of Construction

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## ABSTRACT

Science education in academic engineering programs aims to equip students with scientific knowledge and academic skills to solve complex (socio-)technological problems. This article addresses the critical question of whether traditional science courses effectively prepare for this ability. It starts from the premise that scientific approaches to technological design and development require the ability to construct new scientific knowledge (e.g. scientific models) relevant and adequate to the concrete problem. My central claim is that a dominant view of science, called a *physics paradigm of science*, hinders the effectiveness of science courses. Next, I propose that an alternative view of science, called an *engineering paradigm of science*, is better suited to understanding scientific approaches in technological design and development, and to developing more effective science education. The engineering paradigm of science assumes ‘useful’ knowledge as the goal of science, and thus focuses on the construction of scientific knowledge for epistemic purposes in specific contexts such as technological design and development. The philosophy of science can contribute to a better understanding of the epistemic strategies in knowledge construction. I propose to call this domain of study *the context of construction*.

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Cognitive sciences; Scientific  
models; Model construction

## Introduction

### *Science education in academic engineering programs: aims and claims*

Academic (i.e. university level) engineering programs, such as BSc and MSc programs at Technical Universities in Europe, aim to educate future engineers for academic and professional roles in solving complex (socio-)technological problems. An important learning objective described in policy documents is that academically trained engineers are able to use scientific approaches. There are different opinions about what constitutes a scientific approach.<sup>1</sup> This article focuses on the ability to apply scientific knowledge and methods to problem-solving tasks in technological design and development. For example,

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<sup>1</sup> E.g., Meijers (ed.), *Philosophy of Technology and Engineering Sciences*.

Meijers *et al.* consider the ability to acquire new knowledge and new insights in a goal-oriented methodological manner fundamental to any academic program, including those in engineering.<sup>2</sup>

Science education in these programs usually consists of science courses considered fundamental to the engineering discipline, such as Newtonian mechanics, thermodynamics, fluid dynamics, and electricity and magnetism. Educational programs assume that these courses provide students with relevant scientific knowledge and understanding, scientific methodology, practical methods, and intellectual skills. The critical question is whether these courses effectively contribute to students' ability to use scientific approaches to complex (socio-)technological problems.

The overarching aim of this article is to encourage reflection on this question. I will argue that a traditional philosophical view of science – called a *physics paradigm of science* – obscures our understanding of scientific approaches in technological design and development, especially the use of science in the construction of new knowledge relevant to the problem at hand. Therefore, I propose that an alternative philosophical paradigm – called an *engineering paradigm of science* – is more appropriate, as it emphasizes the *construction of knowledge for specific epistemic purposes*.<sup>3</sup>

My argument combines an analysis of beliefs about the epistemological relationship between science and technology (Section 2), the role of philosophical paradigms of science in ideas about science education (Section 3), and findings in the educational and cognitive sciences on the effectiveness of science education for students' understanding of science (Section 4 and 5). In Section 6, I conclude that science courses in academic engineering programs should focus on the *construction* of scientific knowledge relevant to concrete problems in technological design and development. The philosophy of science can contribute to a better understanding of the strategies and ways of thinking and reasoning in knowledge construction. I propose to call this domain of philosophical study *the context of construction*. The remainder of this section first presents the broader context and relevance of an alternative paradigm of science to science education developed in previous work, and then outlines the approach to taking it further in this article. Table 1 summarizes the structure of my argument.

### ***The ability to construct new scientific knowledge in technological design and development***

My critical reflection starts from the premise that scientific approaches to technological design and development require the ability to construct new scientific knowledge relevant and adequate to the concrete problem. An example is the construction of scientific models of envisioned technological solutions. Academically trained engineers construct these scientific models as 'epistemic tools'<sup>4</sup> that enable their reasoning in the technological

<sup>2</sup> Meijers *et al.*, *Criteria for academic bachelor's and master's curricula*. Also see ABET, "Criteria for accrediting engineering programs."

<sup>3</sup> Boon, "An Engineering Paradigm in the Biomedical Sciences: Knowledge as Epistemic Tool."

<sup>4</sup> Boon & Knuuttila, "Models as Epistemic Tools" and Knuuttila & Boon, "How do Models give us Knowledge? The Case of Carnot's Ideal Heat Engine."

development process. Crucial to my argument is that scientific knowledge (e.g. conceptual and mathematical models to explain, predict, generate, or calculate – naturally or technologically produced – phenomena such as technologically functional properties or processes) is not constructed on the basis of an algorithm, but consists of skillfully collecting and integrating relevant scientific theories, knowledge, and data, usually from several scientific disciplines and using appropriate scientific methodologies.<sup>5</sup>

### ***Engineers as scientist-practitioners***

Miedema introduces the term *scientist-practitioner* to underline that academically trained engineers<sup>6</sup> take scientific approaches in their professional practices. They have the academic and professional skills to analyze problems systematically, translate scientific findings to practical settings, and deal in a scientifically appropriate manner with evidence (e.g. data or knowledge from science or practice). These professionals have acquired relevant scientific knowledge and can design research for questions raised in engineering practices. Therefore, the notion of scientist-practitioners expresses that academically trained engineers combine a *scientific* approach with an *engineering* approach in dealing with (socio-)technological problems that are usually complex.<sup>7</sup> In this educational view, scientific approaches to concrete problems typically concern *constructive* activities such as constructing knowledge (e.g. scientific models), experimental set-ups, and designs or design concepts (i.e. concepts that grasp a specific functionality). Altogether, engineers need to possess flexible and high-level cognitive abilities to apply, develop, and improve specific technologies in broader (socio-)technological settings by using relevant scientific knowledge and methodologies.

### ***Applying science according to the engineering paradigm of science***

Common views on the role of scientific theories in technology resonate with traditional philosophical views about science – called a *physics paradigm of science*. This paradigm is problematic regarding our understanding of how science is used in problem-solving contexts, especially when it comes to constructing new knowledge for a specific epistemic purpose. An alternative philosophical paradigm – called *an engineering paradigm of science* – is a better fit with current *engineering science* practices.<sup>8</sup> Although this alternative is operative in many current research practices, it remains mostly implicit – that is, when scientific researchers, teachers, and science policymakers talk *about* science, they still do so primarily in terms of the traditional physics paradigm.

The engineering paradigm of science emphasizes that scientific research aims to construct ‘useful’ knowledge (e.g. empirical laws, descriptions of phenomena, scientific

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<sup>5</sup> Boon, “Scientific Methodology in the Engineering Sciences.”

<sup>6</sup> As in Meijers et al., *Criteria for academic bachelor's and master's curricula* and ABET, “Criteria for accrediting engineering programs.”

<sup>7</sup> Miedema, *Arts en Ingenieur: And Ever the Twain shall Meet: Analyse en Ontwerp van de Opleiding Technische Geneeskunde*.

<sup>8</sup> Boon, “An Engineering Paradigm.”

concepts, scientific models, and design concepts)<sup>9</sup> and instruments relevant to concrete problems. Instruments can be of different kinds, such as *technological* instruments used in experimentation and measurement to generate data and phenomena,<sup>10</sup> and *mathematical* (e.g. formula and abstract theories) and *cognitive* instruments (e.g. scientific concepts, models, graphs, and diagrams) to enable scientific reasoning. Furthermore, it stresses that the construction of knowledge and instruments for concrete problems occurs within specific disciplinary and problem contexts – proper scientific understanding requires taking these disciplinary contexts into account.<sup>11</sup> All this implies that understanding existing scientific knowledge requires understanding the *context of its construction*, i.e. explaining how researchers originally constructed a specific piece of scientific knowledge (e.g. a concept or model) and instruments. Various aspects of the initial problem – and research-context play a crucial role in this explanation. For example, the articulation of the research problem (e.g. how researchers in a specific discipline frame the practical problem); the (technologically-produced) phenomenon that the theory or model aims to explain; the experimental and measurement techniques used to investigate the phenomenon; the existing scientific (theoretical and empirical) knowledge and concepts relevant to the problem; and the availability of mathematical, technological, and conceptual instruments. Accordingly, ‘applying science’ interpreted from an *engineering paradigm of science* means: making effective use of scientific knowledge, methods, instruments, and ways of thinking – e.g. different kinds of scientific reasoning in problem-analysis, creative and critical thinking, conceptualizing, hypothesizing, mathematizing, modeling, and testing – in scientific approaches to complex (socio-)technological problems.

### ***Science education: a physics versus an engineering paradigm of science***

Science courses in the engineering sciences are still primarily guided by a traditional view of science. Academic engineering education programs, for example, emphasize understanding theories through mathematical modeling. Although mathematical modeling is crucial in the engineering sciences, it covers only a limited part of scientific approaches to new knowledge construction. Yet, a focus on mathematical derivation of models from fundamental theories reinforces the traditional physics paradigm of science by assuming that scientific knowledge about concrete technological phenomena or systems can simply be derived from fundamental scientific theories.<sup>12</sup> Firstly, these kinds of educational approaches assume that understanding the epistemological relationship between scientific knowledge and a concrete phenomenon or system is somehow straightforward, suggesting that applying science to concrete problems is not worth explaining.<sup>13</sup> Secondly, these approaches assume that students learn to *understand* these theories by deriving mathematical models from fundamental theories through exercises in their textbooks. However, little is usually said about how these models link to complex concrete

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<sup>9</sup> Boon, “Scientific Methodology.”

<sup>10</sup> Boon, “Measurements in the Engineering Sciences: An Epistemology of Producing Knowledge of Physical Phenomena.”

<sup>11</sup> Boon & Van Baalen, “Epistemology for Interdisciplinary Research – Shifting Philosophical Paradigms of Science.”

<sup>12</sup> Channell, “The Emergence of the Engineering Sciences: An Historical Analysis.”

<sup>13</sup> Cartwright, “How do we Apply Science?” and *How the Laws of Physics Lie*. Also see Boon, “How Science Is Applied in Technology.”

(technologically-generated) phenomena or systems.<sup>14</sup> The critical question is whether these educational approaches lead to the required academic skills of scientifically-trained engineers.<sup>15</sup>

### ***Against the Received View – alternative views on science education***

To clarify the kind of understanding that future engineers must acquire, the challenge of learning to understand science is considered from an educational angle. For this purpose, Section 4 harks back to scientist-educators in the 1950s and 1960s who criticized the *Received View* of science – i.e. the then-dominant logical-positivist and empiricist tradition that influenced how science is portrayed in science education. Contrary to the *Received View* held by philosophers of science in those days, these scientist-educators emphasized the constructive contributions made on the one hand by interactions with experimental set-ups and technological instruments, and on the other hand by the human cognitive apparatus, which includes not only logical and algorithmic capabilities, but also creative capacities such as to ‘see’ structures, relationships, and analogies, to imagine, conceptualize, and mathematize, and to transform observations into new concepts.<sup>16</sup> Hence, these authors argued that the *Received View* is incorrect about scientific practice and thus should not guide science education. As an alternative, they sought to develop ideas and learning objectives that would be better suited to concrete scientific practices. However, the way science is taught in academic education has not changed much since then. Therefore, I will suggest that the work by these scientists-educators is still valuable for developing a philosophical underpinning of educational views that are more consistent with the constructive character of science assumed in the engineering paradigm of science.

### ***Promoting scientific understanding according to the cognitive sciences***

In Section 5, insights from the cognitive sciences suggest a way forward in science education that agrees well with the engineering paradigm of science. I adopt the suggestion that science education should work from the concrete to the abstract. Rather than just focusing on understanding *abstract* theories as a primary learning objective, science education should emphasize the crucial roles of technological, mathematical, and cognitive instruments used to investigate and interact with aspects of the world to generate (new) knowledge and (new) instruments. Furthermore, it should explain that the human cognitive apparatus potentially possesses unique capacities to perform many different types of thinking and reasoning essential to scientific research. These capacities make it possible for humans to create, construct and invent (scientific) knowledge and other artifacts. Science education in academic engineering programs should promote these capacities in scientist-practitioners,<sup>17</sup> enabling them to competently apply science in constructing *new* knowledge for solving concrete socio-technological problems.

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<sup>14</sup> Boon, “In Defense of Engineering Sciences: On the Epistemological Relations between Science and Technology” and “Scientific methodology.”

<sup>15</sup> As in Meijers et.al., *Criteria for academic bachelor’s and master’s curricula* and ABET, “Criteria for accrediting engineering programs.”

<sup>16</sup> Also see Hanson, “The logic of discovery” and “Seeing and Seeing As.”

<sup>17</sup> Miedema, (in Dutch) *Arts en Ingenieur*.

## Overview

**Table 1.** Structure of the argument

**Premise 1:** Solving complex (socio-)technological problems requires the ability to take scientific approaches in technological design and development, which involves the ability to construct new scientific knowledge (e.g. empirical laws, descriptions of phenomena, scientific models, design concepts) relevant and adequate about the concrete problem.

**Premise 2:** Science courses in academic engineering programs aim to equip students with scientific knowledge and academic skills to promote the ability referred to in Premise 1.

**Claim 1:** A dominant view of science, called a *physics paradigm of science*, hinders the effectiveness of science education in academic (university level) engineering education to develop the ability referred to in Premise 1.

**Claim 2:** An alternative view of science, called an *engineering paradigm of science*, is better suited to understand scientific approaches in technological design and development, and therefore also to develop more effective science education to promote the ability referred to in Premise 1.

|                                                                                                                                                                | Physics paradigm of science                                                                                                                                                                                                                                                                                                                                             | Engineering paradigm of science                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                | <b>Characterized by</b> (see Section 3)                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>The aim of science is:</b>                                                                                                                                  | 'true' knowledge <ul style="list-style-type: none"> <li>knowledge is understood as a truthful, more or less literal representation of aspects of the world.</li> </ul>                                                                                                                                                                                                  | 'useful' knowledge, <ul style="list-style-type: none"> <li>knowledge is also understood as an <i>epistemic tool</i> that allows for reasoning about a problem.</li> </ul>                                                                                                                                                                                                                                                                                             |
| <b>Epistemology – How knowledge is generated:</b>                                                                                                              | <ul style="list-style-type: none"> <li>There is no logic or methodology for generating new knowledge (CoD).</li> <li>Focus on evidencing scientific knowledge (CoJ).</li> <li>Objectivity &amp; rationality is warranted by mere deductive and inductive reasoning.</li> <li><i>Context of Discovery</i> (CoD) versus <i>Context of Justification</i> (CoJ):</li> </ul> | <ul style="list-style-type: none"> <li>The construction of knowledge is a creative process that involves rational <i>epistemic strategies</i> and methods.</li> <li>Focus on understanding how knowledge is constructed.</li> <li>Constructivist (Kantian) epistemology, assuming the indispensable creative contributions through human cognition and technological instruments to the construction of knowledge.</li> <li><i>Context of Construction</i></li> </ul> |
| <b>Science education in academic engineering programs through the lens of these paradigms of science</b>                                                       |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>The epistemological relationship between science and technology:</b> (Section 2)                                                                            | Modern technology is always based on science, which implies that technology derives from fundamental scientific knowledge. <sup>18</sup>                                                                                                                                                                                                                                | Technological thinking has been crucial to the development of science in different ways (e.g. concepts, instruments, perception, measurement, experimentation).                                                                                                                                                                                                                                                                                                       |
| <b>New scientific knowledge (e.g. scientific models) relevant and adequate about the concrete problem in technological design and development:</b> (Section 2) | is deductively or inductively derived from scientific theories produced in science.                                                                                                                                                                                                                                                                                     | involves skillfully collecting and integrating relevant scientific theories, knowledge, and data, usually from several scientific disciplines and using appropriate scientific methodologies.                                                                                                                                                                                                                                                                         |
| <b>Science education focuses on:</b> (Section 4 & 5)                                                                                                           | understanding scientific theories                                                                                                                                                                                                                                                                                                                                       | understanding how to use science in the construction of scientific knowledge                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Educational approach:</b> (Section 4 & 5)                                                                                                                   | Starts from abstract theories to concrete instances that exemplify the theory. (Section 5)                                                                                                                                                                                                                                                                              | Starts from concrete problems for which new knowledge (e.g. scientific models) must be constructed:<br>(a) Abstract theories, scientific laws and concepts are explained by starting from concrete phenomena based on which scientific knowledge was constructed or 'discovered' <sup>19</sup> (i.e. a history of science approach). (Section 4)                                                                                                                      |

(continued)

**Table 1.** Continued.

|                                                                                                                                  |                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Domain of study for the philosophy of science and possible contributions to more effective science education:</b> (Section 6) | <i>Context of Justification (CoJ)</i><br>Analyzing, articulating, and critically assessing the quality of scientific evidence. | (b) Abstract theories, scientific laws and concepts are utilized as perspectives (i.e. as epistemic tools) to ‘describe,’ interpret, and explain the problem – thus generating new knowledge. (Section 5)<br><br><i>Context of Construction (CoC)</i><br>Analyzing, articulating, and critically assessing the epistemic strategies and the wide range of reasoning and thinking processes used in scientific research. |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

<sup>18</sup>E.g., Casimir, *Haphazard Reality - Half a Century of Science*, 295, 297.

<sup>19</sup>Cartwright, *Natures Capacities and Their Measurement*.

## The relationship between science and technology

### *The belief that technology results from science conveyed in science education*

Science courses in academic engineering education often exhibit the belief that technology derives from fundamental scientific knowledge. Courses in thermodynamics, for example, introduce the fundamental laws of thermodynamics and then show how by these equations, one can deduce a mathematical model of the *ideal heat engine*. This educational approach suggests that the mathematical model derived from fundamental laws formed the basis for the invention of the steam engine. However, the father of thermodynamics, Sadi Carnot, laid the foundation for thermodynamics only a century after engineers developed the first steam engine. Newcomen’s first steam engine design is dated 1712, whereas Carnot published his scientific work in 1824.<sup>20</sup>

Carnot conducted his scientific work to understand the fundamental limits of the amount of power that can be produced from heat. In this regard, his work can be considered an example of engineering sciences. He developed his theory based on the actual workings of existing steam engines. As a military engineer, he was motivated by a concrete problem related to this technology, namely, the *efficiency* of steam engines. Engineers already aimed to optimize the steam engine by technological interventions. Carnot’s approach can be regarded as scientific because he abstracted from the technological workings by phrasing the problem more fundamentally:

The question whether the motive power of heat [i.e. the useful effect that an engine is capable of producing] is limited or whether it is boundless [i.e. whether a *perpetuum mobile* is possible in principle] has been frequently discussed. Can we set a limit to the improvement of the heat-engine, a limit which, by the very nature of the things, cannot in any way be surpassed? Or conversely, is it possible for the process of improvement to go on indefinitely?<sup>21</sup>

Thus, unlike how thermodynamics is usually introduced in science courses,<sup>22</sup> the beginnings of fundamental thermodynamic theory resulted from thinking abstractly (i.e. scientifically) about an already existing machine.<sup>23</sup> This example illustrates how generally

<sup>20</sup> Carnot, *Reflections on the Motive Power of Fire, and on Machines Fitted to Develop that Power*.

<sup>21</sup> Carnot, *Reflections*.

<sup>22</sup> Examples can be found on YouTube, such as this Yale lecture: 23. The Second Law of Thermodynamics and Carnot’s Engine <https://www.youtube.com/watch?v=DeNBWsZHXTE>

<sup>23</sup> See Boon & Knuuttila, “Models as Epistemic Tools in Engineering Sciences: A Pragmatic approach” and Knuuttila & Boon, “How do models give us knowledge?”



accepted beliefs about science and technology often determine how we think and talk *about* research practices.<sup>24</sup> These beliefs also motivate and often influence how we teach science and technology.

### ***The roles of technology in the development of science***

The inadequacy of the belief that technological development is always based on science has also been demonstrated in empirical studies in the history of science and technology. Various authors even claim the opposite, namely that technology precedes science. To clarify this in a way that contributes to a better understanding of the consequences of generally accepted philosophical beliefs about the relationships between science and technology for science education, I will use the studies of David Channell.<sup>25</sup> Concerning these beliefs, he writes:

One of the oldest, and probably most common, ideas is the assumption that technology is *dependent* upon science. Since at least the second half of the nineteenth century, there has been the widespread belief, particularly among scientists and the public at large, that *technology simply is applied science ... Science is seen as a precondition for modern technology. ...* On the other hand, *technology simply applies scientific theories and methodologies to practical problems without contributing to, or transforming in any way, that scientific knowledge.*<sup>26</sup>

To counter this complex of interconnected beliefs, Channell claims that *technological thinking* has been crucial to the development of science in different ways throughout history.<sup>27</sup> His examples show that technology and *technological thinking* play different kinds of roles in science. Firstly, technological concepts often precede scientific concepts. Secondly, newly developed technological instruments, such as the mechanical clock, can provide metaphors to promote new metaphysical beliefs about what the world is or how it works. Thirdly, technological instruments such as the telescope and microscope enhance the human senses.<sup>28</sup> Fourth, technological instruments such as the air pump allow experimentation and (quantitative) measurements. Fifth, technological instruments produce reproducible phenomena such as electromagnetism, emission spectra, and piezo-electricity. These phenomena can be investigated in scientific research – for example, how to utilize them for specific technological functions. Sixth, in this latter role, science becomes instrumental to technology development. Altogether, the development of science cannot be adequately understood when neglecting technology's conceptual and instrumental contributions and vice versa. I suggest that these roles of technology in scientific research can be better explained from an engineering paradigm of science. With that in mind, I will propose that students' understanding of scientific theories could be promoted if these roles of technology in scientific research were part of science teaching in academic engineering education.

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<sup>24</sup> Boon, "In Defense of Engineering Sciences" and "How Science Is Applied."

<sup>25</sup> Channell, "The Emergence of the Engineering Sciences."

<sup>26</sup> Channell, "Technological Thinking in Science," 27–8, my emphasis.

<sup>27</sup> Channell, "Technological Thinking in Science," 2015.

<sup>28</sup> Also see Chang, *Inventing Temperature: Measurement and Scientific Progress* and *Is Water H<sub>2</sub>O? Evidence, Realism and Pluralism*.

## Philosophical paradigms of science

### *A matrix for analyzing philosophical paradigms of science*

Our images of science and expectations of what science is capable of can be analyzed in terms of *philosophical paradigms of science*.<sup>29</sup> Thomas Kuhn introduced the notion of paradigms in science.<sup>30</sup> Physicists and philosophers of science heavily disputed his idea about the indispensable role of paradigms giving direction to science because it undermined a cherished image of science according to which science is objective, rational, and anti-metaphysical. Half a century later, most of us have got used to the idea that paradigms play an indispensable role in science, especially, in the way knowledge is constructed and justified. In current usages of the term in philosophy and other academic fields, it is often used in a less rigid form than suggested by Kuhn, thus assuming that paradigms can co-exist.

A paradigm is loosely understood as a kind of ‘image,’ ‘picture,’ a ‘frame,’ or a ‘web of beliefs’ that works in the background when we think and talk about problems and solutions. Usually, we are not aware of its workings, such as *how* we perceive a problem and *how* we think about solving it. Currently, it is generally assumed, also based on research in the cognitive sciences, that this kind of background is crucial in every knowledge practice.<sup>31</sup> So-called *disciplinary paradigms* enable researchers within a specific discipline to make sense of observations and data, scientific knowledge and concepts, and to justify their scientific reasoning methods. Therefore, the disciplinary paradigm directs or even determines, usually in an implicit manner, how researchers phrase a problem and conceive of possible solutions.<sup>32</sup>

In the second edition of his ground-breaking work – after being criticized for the imprecise meaning of the notion – Kuhn specified the content of a paradigm by a so-called *disciplinary matrix*. The disciplinary matrix distinguishes between four crucial aspects of paradigms in science: symbolic generalizations, metaphysical presuppositions, values to judge a theory, and exemplars.<sup>33</sup> Thus, Kuhn’s clarification suggests that the content of a paradigm can be analyzed and articulated by specifying the elements put forward in the disciplinary matrix.

Building on Kuhn’s approach, I have proposed a matrix for articulating philosophical paradigms of science, i.e. a matrix to analyze the background images(s) (i.e. the paradigm) we maintain *about* science.<sup>34</sup> The matrix distinguishes between thirteen elements that constitute a philosophical paradigm of science. This matrix thus provides a framework for articulating the distinct elements of the paradigm. Moreover, the specific content of the elements in the matrix are mutually coherent and as an integrated coherent whole they form a specific paradigm (i.e. an image of science). It is important to recognize that a paradigm is thus a *construct* rather than an empirically correct description of peoples’ views on science.

<sup>29</sup> Boon, “An Engineering Paradigm.”

<sup>30</sup> Kuhn, *The Structure of Scientific Revolutions* [1962].

<sup>31</sup> Also see Andersen & Nersessian, “Nomic Concepts, Frames, and Conceptual Change.”

<sup>32</sup> Boon & Van Baalen, “Epistemology for interdisciplinary research.”

<sup>33</sup> Kuhn, *The Structure of Scientific Revolutions* [1970], “Reflections on my critics,” and “Second thoughts on paradigms.”

<sup>34</sup> Boon, “An Engineering Paradigm.”

Using this matrix, two mutually contrastive paradigms have been constructed, labeled a *physics paradigm of science* versus an *engineering paradigm of science*.<sup>35</sup> The first reflects a *scientific realist* philosophy of science and fits with beliefs in the basic sciences, such as physics and chemistry. The engineering paradigm of science reflects a *pragmatist* philosophy of science<sup>36</sup> and an epistemologically motivated constructivism,<sup>37</sup> which is better suited to scientific research practices that aim to contribute to concrete real-world problems, such as the engineering sciences.

### ***A physics paradigm of science implicitly guides science education***

Many science courses accord with (and convey) the beliefs inherent in the physics paradigm of science. For example, the belief that higher-order phenomena can be explained by knowledge about more fundamental phenomena. Even if scientific practices show that this approach does not always work well, science teachers are still used to teaching it in this way. This choice may be promoted by the belief articulated in the physics paradigm that the fundamental sciences provide knowledge that ‘in principle’ can be applied in the engineering sciences. This belief provides a rationale for teaching science and scientific research in the ways we usually do. As will become apparent below, critical ideas from science education and insights from the cognitive sciences in combination with the engineering paradigm of science provide starting points for alternative educational approaches.

### ***Scientific research according to an engineering paradigm of science***

Contrary to the physics paradigm with its focus on fundamental theories, the engineering paradigm encompasses the idea that the *epistemic aim of science* (i.e. matrix element I in the engineering paradigm) is to produce knowledge that allows for epistemic uses, such as in problem solving, i.e. the relevant and adequate knowledge for intended epistemic uses (e.g. the epistemic uses of a scientific model in concrete problem-solving). This idea is also reflected in the *epistemic and pragmatic values and criteria for accepting knowledge* (matrix element II in the engineering paradigm). Examples are values and criteria such as: empirical adequacy, reliability and relevance, given epistemic purposes that concern practical uses; simplicity in the sense of manageability and tractability; a balance between generality and specificity given epistemic aims; explanatory strength; predictive reliability; logical consistency; coherence with accepted knowledge relevant to the epistemic uses; integration of knowledge from different fields and levels; and validation in regard of intended epistemic uses.

Altogether, the matrix elements that constitute the engineering paradigm ultimately focus on how scientific research yields *useful* knowledge, that is, useful for all kinds of epistemic tasks, such as *explaining, calculating, predicting, simulating, and thinking creatively* about a concrete problem or design concept. Unlike the physics paradigm, this implies that research *topics* are not primarily justified by metaphysical beliefs (i.e. the belief that the purpose of science is to discover the basic structure or ‘furniture’ of the world). Moreover,

<sup>35</sup> Ibid.

<sup>36</sup> Chang, “Pragmatism, Perspectivism, and the Historicity of Science.”

<sup>37</sup> Boon, “Philosophy of Science in Practice: A proposal for Epistemological Constructivism.”

research *methodologies* are not firstly motivated by metaphysical or ontological beliefs. Reductionist methodologies are still used, but the rationale for such a choice is based on practical and pragmatic considerations about the most effective approach in a specific case. Fundamental theories also play a role. But according to the engineering paradigm, those theories are firstly interpreted as theoretical perspectives,<sup>38</sup> ‘knowledge instruments,’ or ‘tools for thinking,’ or ‘epistemic tools,’ for example when a theory used as a theoretical perspective for constructing a model of concrete systems. Therefore, pragmatic and epistemological considerations play a crucial role in the many different types of choices made during the research process, where the expertise of the researcher is to adequately weigh and assess various options.

## Practice-based ideas about science education

### ***What should be taught in science education? The nature of science (NOS), according to James T. Robinson***

James T. Robinson was one of the first authors to ask the kind of critical question posed in this article. His work concerns secondary education, but I suggest that it also concerns science teaching in academic engineering education. Moreover, his account of the nature of science is relevant to my argument because it can be considered a forerunner of the engineering paradigm of science concerning science education – this motivates the extensive outline of his ideas below.

Robinson’s research into the mentioned question – culminating in a 1964 Ph.D. thesis<sup>39</sup> at Harvard University – was motivated by two concerns that occurred to him as a practicing high-school biology teacher:

First, the growth of knowledge was exacerbating the problem of choosing the appropriate knowledge, skills and understandings central to the sciences and fundamental enough to enable citizens to continue learning science as knowledge changed ... Second, most public school texts organized the sciences into separate topics that failed to interrelate principles, laws and theories within which these topics gained their significance.<sup>40</sup>

The first problem pointed out in this quote has only become worse. The number of scientific disciplines continues to grow, and, in its wake, the number of BSc and MSc programs. The second problem concerns the separation of scientific disciplines, which leads to increasing fragmentation. The effect is that professionals’ ability to use science in complex problem-solving tasks is hindered because bridging dividing lines and establishing relations between scientific disciplines is not usually part of educational programs.<sup>41</sup> These are significant issues with which the current education of professionals (i.e. scientist-practitioners) and academics has to deal.

Robinson proposed that students should also learn *what* science is, which he dubbed as understanding the *nature of science* (NOS). According to him, science education in high school aims at *scientific literacy*, which requires a minimal understanding of the nature of science, i.e. an understanding of the *prevalent philosophical view of science*, including insight

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<sup>38</sup> Giere, *Scientific Perspectivism*.

<sup>39</sup> Robinson summarized his thesis (1964) in an article: “Science Teaching and the Nature of Science.”

<sup>40</sup> Robinson, “Science Teaching,” 635. Also see Robinson, *The Nature of Science and Science Teaching*.

<sup>41</sup> Also see Boon & Van Baalen, “Epistemology for Interdisciplinary Research.”

into how scientific knowledge is constructed. Therefore, crucial to Robinson's view is that students should learn not only the *products* of science but also the *processes* by which scientific knowledge is produced.

### ***Intended learning objectives of teaching NOS in science education***

In his attempt to develop an understanding of the nature of science (NOS) 'from within,' Robinson first turned to the philosophy of science literature of his days, in particular, logical empiricism, but found that professional philosophy was rather technical and "left out a lot of the work of the scientist." He reverted to studying "works of scientists who had turned to philosophy and had written about the philosophy of science from their perspectives as contributors to original scientific research."<sup>42</sup> Robinson chose significant philosophical works by physical scientists and biologists.<sup>43</sup>

To indicate what kind of understanding a scientifically literate person should acquire, Robinson proposed a list of 85 propositions. At present, we would call these propositions *intended learning objectives*. Many of these propositions are still worth considering as they focus on the intellectual, epistemological, and methodological aspects of the *processes* of knowledge production. Some telling examples of these propositions (understood as intended learning objectives) that fit well with a vision of science education in academic engineering programs based on an engineering paradigm of science are:

- *Understand* how changes in metaphysical principles can bring about profound changes in the structure of scientific knowledge.
- *Understand* the inextricable relationship of the knower and the known.
- *Understand* the impossibility of divorcing physical concepts from the operations by which they are generated,<sup>44</sup> and the impossibility of speaking of things existing by themselves in their own right.
- *Understand* the relationship of theory to observation – without theory, man does not know what to observe.
- *Understand* the use of imagination and abstraction, which are essential characteristics of the processes of discovery in science.

Robinson's nature of science (NOS) is significantly different from logical positivism and empiricism promoted in the philosophy of science of his days.<sup>45</sup> Robinson's difficulty with the academic logical positivist and empiricist philosophy of science at that time, and his choice to instead study the philosophical writings by scientists, make sense when we recognize that he sought to understand the nature or structure of scientific knowledge in a way that is useful for science education. As said, according to Robinson, science education must include the processes of science, for example, how observations and data are converted into scientific knowledge, including the concepts and principles with which the products of science are created.<sup>46</sup> Robinson's approach to the NOS was at odds with most of the

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<sup>42</sup> Robinson, "Science Teaching," 636.

<sup>43</sup> Also see Matthews, "James T. Robinson's Account of Philosophy of Science and Science Teaching: Some Lessons for today from the 1960s" for a comprehensive overview of Robinson's ideas and its reception.

<sup>44</sup> Also see Bridgman, "Operational Analysis" and Chang, "Operationalism."

<sup>45</sup> See for example, Frederick Suppe, "The Search for Philosophical Understanding of Scientific Theories."

<sup>46</sup> Also see, Polanyi, *Personal Knowledge: Towards a Post-Critical Philosophy*, and *The Tacit Dimension*; Schön, *The Reflective Practitioner – How Professionals Think in Action*; Toulmin, *The Uses of Argument and Return to Reason*.

academic philosophy of science, which, after all, tried to avoid any reference to the role of the researcher in how the products of science were created. Philosophers focused on the *context of justification*, deliberately ignoring the *context of discovery*. Conversely, Robinson chose philosophical works of scientists who explain how scientific knowledge is created, and who thereby reveal the constructive character of, as well as the role of metaphysical assumptions underlying, scientific inquiry.<sup>47</sup>

Important for my argument is that the focus on the justification of theories in the traditional philosophy of science suggests that knowledge is ultimately based on observations and measurements. For example, to justify the objectivity and rationality of science, the *formation of scientific concepts* needs to be reduced to a kind of inductive inference or results from spontaneous ‘flashes of inspiration’ in scientific discovery. Despite philosophical criticism,<sup>48</sup> the role of human cognition, creativity, and experimentation in knowledge construction is still neglected, for example when explaining scientific methodology. Critics such as Robinson, with interest in the processes of science, on the other hand, emphasize the indelible role of constructive activities. Yet, whereas the engineering sciences constantly create new concepts, students are rarely made aware of these creative knowledge production processes.

Robinson’s ideas about science education<sup>49</sup> and *understanding science* – i.e. what it takes to understand the nature of science as expressed in his 85 propositions – may be too advanced for high-school-level students. Here, I suggest that his views are still valuable for our thinking about science education at the academic level – especially when we strive for students to understand the research process, i.e. understand *how* to conduct scientific research and construct scientific knowledge. It emphasizes the indispensable roles of human cognition and imagination, the unavoidable role of metaphysical principles, and the essential roles of existing theories and concepts together with technology and experimental procedures in discovering scientific concepts and theories that enable our epistemic interactions with the world.

### **James B. Conant on the significance of teaching the history of science**

A complementary approach to better understanding science is teaching the *history* of science. One of the first proponents of this pedagogical approach was the chemist and president of Harvard, James B. Conant. Conant was concerned with the pedagogical problem of how science teaching could give “a better understanding of science to those of our graduates who are to be lawyers, writers, teachers, politicians, public servants, and businessmen.”<sup>50</sup> Not every student aspires to be a scientist. Most of them are educated for professional roles. Moreover, says Conant, no student has the ability to acquire an understanding of all the relevant scientific knowledge, let alone an understanding of how to use this vast knowledge in assessing all the issues that concern their duties as a citizen in a

<sup>47</sup> Robinson’s “An Investigation of Selected Frameworks of Science,” 1964 complies in many respects with Kuhn, *The Structure of Scientific Revolutions* [1962]. Also see Robinson “Philosophy of Science: Implications for Teacher Education” and “Philosophical and Historical Bases of Science Teaching.”

<sup>48</sup> For example, Nersessian, *Creating Scientific Concepts*; Rouse, “Articulating the World: Experimental Systems and Conceptual Understanding”, and *Articulating the World: Conceptual Understanding and the Scientific Image*; Feest, “Concepts as Tools in the Experimental Generation of Knowledge in Psychology”; Boon, “Scientific Concepts.”

<sup>49</sup> For example, Robinson, *ibid.*

<sup>50</sup> Conant, *On Understanding Science: An Historical Approach*, 17.

democratic society. In the current context, the same can be said about academically-trained engineers acting in their professional capacity (i.e. as scientist-practitioners). Some will choose careers in the engineering sciences at university or industrial research institutions to work on technological development and innovation. Others will work on technology development within the corporate world. However, for the creative, effective, and responsible (e.g. in the sense of safety, reliability,<sup>51</sup> and sustainability) development of new technology, it is crucial to use scientific insights and research.

Understanding science, according to Conant, requires some research experience:

Even a highly educated and intelligent citizen without research experience will almost always fail to grasp the essentials in a discussion which takes place among scientists concerned with a projected inquiry. This will ... be to a large degree because of his fundamental *ignorance of what science can or cannot accomplish* ... He has no 'feel' for the *Tactics and Strategy of Science* ... *Being well informed about science is not the same thing as understanding science*, though the two propositions are not antithetical. *What is needed are methods for imparting some knowledge of the Tactics and Strategy of Science to those who are not scientists* ... [In this manner] enough can be accomplished, I dare hope, to bridge the gap to some degree between those who understand science because science is their profession and those who have only studied the results of scientific inquiry – in short, the laymen.<sup>52</sup>

Thus, understanding science is not achieved by more scientific information but by learning how scientific researchers approach problems, i.e. their *tactics* and *strategies*. Conant, therefore, defends a historical method as a way towards understanding science. Like Robinson, he believes that the philosophical analysis of science and its methods has led to a misunderstanding about science because the philosophy of science of his time neglected the crucial roles of experience and experimentation.<sup>53</sup> Authors have been defending history and philosophy of science approaches to science education ever since.<sup>54</sup>

Experiences in experimentation and reasoning processes, according to these authors, help students to better understand science 'from within.' This insight is vital for the academic education of engineers, who need to learn how to construct new knowledge for specific problems (e.g. empirical descriptions of phenomena<sup>55</sup> relevant to their object of study, and scientific models<sup>56</sup> explaining these phenomena). Furthermore, in line with Conant, I defend the position that academic engineering education should aim at learning the *tactics* and *strategies* of scientific researchers in approaching problems – whether this should be done through case histories or some other approach remains an open educational question.

### **Turning focus from understanding theories to understanding how to use science**

Approaches such as teaching the nature of science as suggested by Robinson, the historical method proposed by Conant, and the history and philosophy of science approach

<sup>51</sup> E.g., Douglas, "Inductive Risk and Values in Science."

<sup>52</sup> Conant, *On Understanding Science*, 26–7, my emphasis.

<sup>53</sup> Conant, *Harvard Case Histories in Experimental Science*.

<sup>54</sup> E.g., Matthews, "The History and Philosophy of Science in Science Teacher Education" and *Science Teaching: The Contribution of History and Philosophy of Science*.

<sup>55</sup> E.g., Bogen & Woodward, "Saving the Phenomena."

<sup>56</sup> E.g., Bailer-Jones, *Scientific Models in Philosophy of Science*.



advocated by Michael Matthews, have several advantages. Firstly, it leads to a better understanding of the nature of science as a human endeavor in a societal context. Secondly, it improves students' understanding of scientific theories. Thirdly, students become aware of the ineradicable role of the context within which the theory is 'discovered.' Finally, understanding the crucial roles of contextual aspects – such as human cognition and imagination, metaphysical principles, theories, concepts, technology, experimental procedures, and the (societal and technological) problem-context – may prevent that students take theories in a very literal and rigid fashion.

These early pedagogical approaches to improve the understanding of science are not only valuable for science education on the high-school level – they are also valuable for science education in academic engineering programs because they promote students' ability to explain the theory and to use the theory in their reasoning about problems. However, the presented ideas on what understanding science means are too limited when it concerns the education of professionals in the sense of *scientist-practitioners*. My critical point is that the mentioned authors prioritize understanding *theories* (e.g. the atomic theory, the phlogiston theory, the caloric theory, and the kinetic gas theory, as in Conant<sup>57</sup>) – they assume that *understanding science* and *scientific understanding* firstly means *understanding scientific theories*. This idea about scientific understanding is also still prominent in current philosophy of science, such as Henk De Regt's recent monograph.<sup>58</sup> I contend that exclusively understanding science in this limited manner still runs the danger that students do not learn to understand how the theory connects with the real world. Students learn to reason within the confines of the theory (as in physics exercises) but not how to apply it when they aim to understand a 'real-world' problem in a way that allows for dealing with it.<sup>59</sup>

The proposed pedagogical approaches seem to embrace an essential assumption of the *physics paradigm*, namely the assumption that the acquisition of scientific knowledge and understanding of the theory is sufficient for the ability to apply science in solving concrete problems. However, this level of knowledge and understanding is usually not enough. First, one must know where, why, and how the theory can be applied.<sup>60</sup> Moreover, knowledge about the concrete problem cannot usually be deduced from the theory<sup>61</sup> but requires selecting and integrating relevant scientific and empirical knowledge from different sources and disciplines, which must be put together into a conceptual model of the phenomenon or system to begin with.<sup>62</sup> Therefore, the critical questions to be addressed are: what do students need to learn about the uses of science, and how can science education accommodate students' needs?

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<sup>57</sup> Conant, *Harvard Case Histories*.

<sup>58</sup> De Regt, *Understanding Scientific Understanding*, particularly page 102. Also De Regt & Dieks, "A Contextual Approach to Scientific Understanding," 151, propose the *Criterion for the intelligibility* of a theory (CIT): "A scientific theory T (in one or more of its representations) is intelligible for scientists (in context C) if they can recognize qualitatively characteristic consequences of T without performing exact calculations."

<sup>59</sup> Note that in an engineering paradigm of science, "aiming to understand a problem" usually involves the construction of a conceptual model of the problem (i.e., the construction of new knowledge), as in Boon "Scientific methodology."

<sup>60</sup> Procee, "Reflection in Education: A Kantian Epistemology,"

<sup>61</sup> Cartwright, *How the Laws of Physics Lie*; Also see Boon, "How Science Is Applied."

<sup>62</sup> Nersessian, "How do Scientists think? Capturing the Dynamics of Conceptual Change in Science?," and "How do Engineering Scientists think? Model-based simulation in biomedical engineering research laboratories;" Boon & Knuuttila, "Models as Epistemic Tools;" Knuuttila & Boon, "How do Models Give Us Knowledge?;" Magnani (ed.), *Model-based Reasoning in Science and Technology: Theoretical and Cognitive Issues*; Magnani & Bertolotti (eds.), *Springer Handbook of Model-Based-Science*.



## Philosophical ideas about science education in academic engineering education

### *The role of human cognition*

One of my suggestions for science education in academic engineering education is to explicitly emphasize and explain the role of the human cognitive system, including the role of 'non-empirical' concepts and *regulative principles* that are necessary for the possibility of human reasoning and imagination in generating knowledge from experiential and empirical input.<sup>63</sup> Kant's epistemology has shaped my ideas on this matter (i.e. epistemological constructivism).<sup>64</sup> Paul Ziche presents a clarifying explanation of Kant's ideas about the constructive nature of science, showing that Kant developed an open methodology that is intended to give direction to our cognitive practices without determining their results, i.e. without providing *certainty* about the outcome.<sup>65</sup>

Ziche's interpretation is relevant to our thinking about science education aimed at training professionals who are able to understand and conduct scientific research because it explains how science is both *creative* and *critical* in constructing new knowledge. Accordingly, I interpret human cognition as a system that enables cognitive activities in scientific research such as: *different kinds of reasoning* (e.g. deductive, inductive, analogical, explanatory, hypothetical, and mathematical reasoning); *determinative judgment*, which according to Kant, is at stake when one tries to situate something particular under a given universal concept or idea, also called 'subsumption under a concept'; *reflective judgment*, which according to Kant plays a role when one seeks to bring forward unity about particulars without having at one's disposal a general concept or idea, for example, by 'seeing' analogies and 'borrowing' concepts from other disciplines<sup>66</sup>; and generating scientific concepts through *experimentation*,<sup>67</sup> *imagination*, and *visualization*.<sup>68</sup>

This interpretation of the role of the human cognitive system in science can help in explaining the crucial roles of, firstly, concepts and regulative principles that have no clear empirical basis (but, notably, must fit with our experience of the world), and secondly, critical and creative thinking in science. I suggest that this interpretation offers an acceptable alternative to the traditional belief that science is, or should be, objective and rational in the sense that 'the world speaks for itself.' This traditional belief trivializes the contribution of the human cognitive system, making it very hard in traditional science education to explain and teach the crucial roles of imagination, creativity, conceptualization, and critical reasoning, in the construction of knowledge.

<sup>63</sup> See Boon, "Philosophy of Science in Practice." Also see Chang, "Ontological Principles and the Intelligibility of Epistemic Activities," and *Inventing Temperature*.

<sup>64</sup> Boon, "Philosophy of Science in Practice," 2017.

<sup>65</sup> Ziche, "Epistemic Confidence: Kant's Rationalization of the Principles of Seeking and Finding."

<sup>66</sup> See Procee, "Reflection in Education." Also see Ziche, "Epistemic Confidence."

<sup>67</sup> For example, Feest, "Concepts as Tools;" Boon, "Scientific Concepts in the Engineering Sciences" and "Measurements in the Engineering Sciences."

<sup>68</sup> For example, Hesse, *Models and Analogies in Science*; Nersessian, *Creating Scientific Concepts*; Rouse, "Articulating the World", and *Articulating the World*.

## ***From the concrete to the abstract and back – the cognitive structure of scientific theories***

Ronald Giere argues that science teaching usually starts at the level of axiomatic theories and thus assumes an axiomatic approach to understanding theories. This educational approach includes, according to Giere, that the meaning of scientific concepts is derived from and defined by the axiomatic theory. A typical axiomatic presentation starts with abstract laws such as Newton's laws. Only then is the law of gravity and specific initial conditions added to the explanation, and the description becomes concrete in representing a tangible physical system such as a real pendulum.<sup>69</sup> Giere argues that this path should be reversed by starting from the concrete to the abstract.

Accordingly, Giere defends the importance of *basic level concepts* and *models*, by which he means concepts and models about which we have a common, everyday understanding. Based on insights of the cognitive sciences, Giere claims that this understanding does not derive from the internal structure of the model (i.e. the formal structure derived from the axiomatic theory, for example, the *internal structure* of the model of the pendulum) or the *definition* of a concept, but from various cognitive interactions between the human agent and the real-world systems these models represent. Like scientist-philosophers (Section 4), Giere highlights the role of *activities*<sup>70</sup> in forming and understanding scientific concepts and models. In particular, Giere highlights the *artifactual* character of models. We *do* things with models. Humans have *cognitive interactions* with models, which is how humans gain understanding. Indeed, Giere argues that many famous scientists' discoveries started with experimentation. Experimentation can be understood as a process in which humans simultaneously explore *physical* interactions with real-world systems and *cognitively* interact with the representations of these systems (i.e. with concepts and models). In this manner, they develop an understanding of the world. The theory is the product of this process. Giere, therefore, suggests that understanding the structure of scientific theories should be understood as the result of the research process.<sup>71</sup>

In agreement with Giere, I suggest that explanations of scientific theories in academic science education must begin at the level where people experiment with physical, technological, and cognitive artifacts (such as concepts and models).<sup>72</sup> It is essential to show that experimental interaction with physico-technological artifacts goes hand-in-hand with cognitive processes because researchers want to explain and give meaning to what they observe during the experimental process. Therefore, next to interactions with physico-technological artifacts that generate new observations and experiences, the cognitive interactions of researchers with concepts and models to represent or interpret these experiences are a crucial part of the research process.<sup>73</sup> Experimentation involves a continuous concept formation and modeling process, usually by building on existing cognitive artifacts

<sup>69</sup> Giere, "The Cognitive Structure of Scientific Theories," 294. Also see Van Fraassen, *Scientific Representation*.

<sup>70</sup> Also see Bridgman, *The Logic of Modern Physics* and "Operational Analysis."

<sup>71</sup> Also see Chang, "The Persistence of Epistemic Objects Through Scientific Change," "The Philosophical Grammar of Scientific Practice," and "Pragmatism, Perspectivism."

<sup>72</sup> Also see Hesse, *Models and Analogies*; Cartwright, *How the Laws of Physics Lie*; Nersessian, "How do Scientists think?," Morrison and Morgan, "Models as Mediating Instruments."

<sup>73</sup> Also see Hacking, *Representing and Intervening: Introductory Topics in the Philosophy of Natural Science* and "The Self-vindication of the Laboratory Sciences."

(e.g., previously constructed concepts and models or premature versions thereof).<sup>74</sup> Interactions and interventions with the artifactual occur in different ways that mutually influence each other.

It is in terms of this process that scientific theories must be understood. This means that the resulting structure of the theory is a cognitive structure in the sense that it is consistent, not only with the empirical world but also with how human cognition interprets and structures perceptions.

### **How is science applied?**

One of the concerns in academic engineering education is that students' ability to apply scientific theories to concrete real-world problems remains limited.<sup>75</sup> This educational problem consists of two parts: first, students do not understand the theory, i.e. they do not understand the internal structure of the theory, and as a consequence, they do not understand how to reason *internal* to the theory. Secondly, students do not understand how to apply the theory, i.e. they do not understand how it is connected to the world *external* to the theory, and thus, how they can reason with the help of the theory about the external world. Authors cited in the previous sections emphasize the importance of explaining how the theory has been generated to improve students' understanding of science, for example, by teaching cases in the history of science. I side with these authors because this approach to science education will improve the first problem. However, it is insufficient as a solution to the second.

Giere has proposed a different view of the nature of scientific theories, which I consider fruitful concerning the second problem. As outlined above, Giere proposes understanding the *structure of theories* as *cognitive structures* generated by human actors based on their concrete experiences and activities.<sup>76</sup> Conversely, in science and science education (and in agreement with the physics paradigm of science), we tend to regard abstract theories as more 'real' than concrete experiences. The reversal of Giere, to view experiences as more real than theories, fits well with the Kantian epistemology (outlined above)<sup>77</sup> that is part of the engineering paradigm of science.<sup>78</sup>

This alternative epistemology has severe consequences for how we think about the role of theories. According to Giere, abstract axiomatic theories should not be interpreted as literal *descriptions* or *representations* of what reality 'really' is, but as *perspectives* on the world through which human agents can 'look at' problems, to interpret and organize these problems, and to generate specific types of models that represent aspects of the world in a specific kind of way.<sup>79</sup> Human actors can identify a real-world system as, for example, a *Newtonian system* (e.g. seeing system S as an oscillating object) or as a *thermodynamic system* (e.g. seeing the same system S as a process that converts one form of energy into another form of energy). Then, by using Newton's axiomatic theory – or, when seen as

<sup>74</sup> Hacking, *Representing and Intervening*; Franklin, "The Neglect of Experiment;" Rheinberger, *Toward a History of Epistemic Things*; Radder (ed.), *The Philosophy of Scientific Experimentation*; Hanson (ed.), *The Role of Technology in Science*; Boon, "Measurements in the Engineering Sciences."

<sup>75</sup> Van den Beemt et al., "Interdisciplinary engineering education: A review of vision, teaching, and support."

<sup>76</sup> Giere, "The Cognitive Structure," 294.

<sup>77</sup> Boon, "Philosophy of Science in Practice."

<sup>78</sup> Boon, "An Engineering Paradigm."

<sup>79</sup> Giere, *Science without Laws and Scientific Perspectivism*.

a thermodynamic system, by using the fundamental laws of thermodynamics – humans can generate models representing system S as a Newtonian system or as a thermodynamic system, respectively.

An educational challenge is then learning to use theories as *perspectives* or *tools* to interpret and structure experiences of the world, for example, regarding phenomena or problems or design concepts in a specific discipline. This approach means that students first learn to view observations and experiences as interpretable in terms of a specific theory, and next, *how* to use the theory to interpret these observations and experiences. It also requires learning to use scientific theories in constructing scientific models for phenomena, problems, and design concepts, in ways that are epistemically relevant to the concrete, real-world problem.

### ***How to include the role of technology in science education?***

Section 2 provided an overview of the roles of technology in the development of science. Science education tends to neglect these roles so that theories seem to float in a vacuum. The *physics paradigm* of science suggests that theories are ‘discovered,’ ultimately independent, rather than dependent on the (technological and conceptual) context in which they were constructed. This supports the focus on theories in science education. Additionally, students learn to reason internal to the theory, but it often remains unclear how to connect the theory with experiences of the real world (including technology) external to the theory. However, much of our theories are tied to technological devices because they concern technologically generated phenomena (e.g. the functional properties of materials in nanotechnology or biomedical technologies). This implies that scientific research also aims at scientific knowledge about these advanced technologies, not so much to learn about Nature but to learn, for example, how to create or improve the technology for specific purposes or functions.

Furthermore, technological measurement instruments generate observations and phenomena based on which scientific theories are built. This implies that measurement technology often is the only real-world connection with the variables and parameters in theories. Therefore, concrete experiences and activities on which theories are based also include those related to the physico-technological and cognitive interactions with the measurement technology. In other words, technologically-generated observations and data form an inherent part of the structure of a theory. Insights into the role of the measurement technology in how a theory was constructed will contribute to students’ understanding of both the *internal* structure of the theory and how the theory is linked to the *external* world.<sup>80</sup>

Finally, technology offers new concepts, or even metaphors, in terms of which experiences and phenomena can be interpreted, thereby contributing to the creative part of scientific research (see Section 2). Given the importance of conceptual interpretation, this role of technology should also attain proper attention in science education.

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<sup>80</sup> See also Boon, “Measurements in the Engineering Sciences.”

## The context of construction

The main question addressed in this article is how science should be taught so that academically-educated engineers will be able to apply science (i.e. using scientific knowledge, methods, and research) in constructing (new) knowledge and (new) instruments for solving complex (socio-)technological problems.

My central claim is that a traditional view of science, called a *physics paradigm of science*, hinders the effectiveness of science education in these programs. My argument combines analyses from the philosophy of science with findings in the history of science and technology and the educational and cognitive sciences.

Firstly, I have argued that the physics paradigm supports the assumptions that technology development derives from fundamental scientific theories, and science has discovered the phenomena utilized in technology. According to this paradigm, knowledge is more or less deductively derived from fundamental scientific theories in the so-called applied sciences (such as the engineering sciences). Historians have criticized these assumptions by showing that the development of science and technology is closely intertwined.

Secondly, I claim that the physics paradigm of science steers the way science is taught. Traditional science education focuses on scientific theories and assumes that this knowledge automatically allows for productive applications of science in the development of technology. The physics paradigm of science entails several assumptions about the role of science in generating knowledge for problem-solving tasks, such as: (i) knowledge about concrete problems can be derived from scientific theories in a more or less straightforward manner; (ii) when students understand the internal structure of a theory, they are able to apply it to real-world problems external to the theory, i.e. they understand *where* to apply it and *how* to apply it; (iii) the only roles of experiments and technological instruments are the development and testing of the theory, which implies that scientific theories can be taught without explicitly incorporating those roles; and (iv) sound scientific reasoning consists exclusively of inductive and deductive reasoning. These assumptions legitimize the idea that students should learn fundamental scientific theories that they can apply to the development of technology.

Thirdly, my analysis utilizes educational and cognitive science insights, focusing on students' cognitive difficulties in understanding and applying abstract theories in generating knowledge (e.g. scientific models) for complex problem-solving tasks. Practicing scientists and science teachers have argued that students' understanding is promoted when science education focuses not only on the products of science (e.g. the abstract theories) but also on the research process, where the indispensable creative contributions through human cognition and the interactions of researchers with experiments and technological instruments are explained. These insights agree with the cognitive sciences, which suggest that science education should start from concrete experiences, not abstract theories.

Based on these analyses, I conclude that science education should focus on the construction of scientific knowledge relevant to the concrete problem-solving task. Science education in academic engineering programs should thus promote students' understanding of how new scientific knowledge is constructed, how this knowledge relates to the problem (or design-concept), and how it meets epistemic and pragmatic criteria crucial

to the problem at hand (e.g. relevance, reliability, and intelligibility). I suggest that this approach to science education will better serve the intended learning objectives related to the uses of science in problem-solving tasks.

An *engineering paradigm of science* is proposed as an alternative better suited to underpin the suggested approach in science education. Whereas the physics paradigm focuses on the discovery and justification of theories as the primary aim of science, the engineering paradigm focuses on the construction and epistemic uses of knowledge in the context of problem-solving tasks. Furthermore, contrary to the traditional philosophical distinction between the *context of discovery* and the *context of justification*, which suggests that there is no logic or methodology in constructing new scientific knowledge, the engineering paradigm of science assumes that the construction of knowledge is a creative process that involves rational epistemic strategies and methods. I propose to call this *the context of construction*.

Authors in the philosophy of science have already criticized the strict distinction between the *context of discovery* and the *context of justification*.<sup>81</sup> For example, Theodore Arabatzis argues that “there is overwhelming (historical and empirical) evidence that hypothesis generation and theory construction are reasoned processes whose explication can (and should be) carried out by philosophers of science.”<sup>82</sup> These findings support the idea that the construction of knowledge involves rational epistemic strategies and methods.

Philosophical focus on the context of construction, therefore, opens new possibilities for the philosophy of science to systematically investigate how scientists reason and justify conclusions in actual scientific research.<sup>83</sup> It entails that knowledge-construction processes, epistemic strategies, and reasoning processes can be analyzed, articulated, and critically assessed. The epistemic strategies and reasoning processes encompass not only inductive and deductive reasoning but also analogical, explanatory, and hypothetical reasoning together with thinking processes such as analysis, integration, abstraction, contextualization, and conceptualization.

Taken together, this also implies that the construction of knowledge involves reasoning processes, epistemic strategies, and thinking skills that need to be taught in science education, especially in academic engineering programs.

The turn from a physics paradigm to an engineering paradigm of science, therefore, points to an essential role for the philosophy of science, as it can make a significant contribution to science education in academic programs by analyzing, articulating, and critically assessing the epistemic strategies and the wide range of reasoning and thinking processes used in scientific research. This domain of study for the philosophy of science is thus called *the context of construction*.

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<sup>81</sup> e.g., Schickore & Steinle (eds), *Revisiting discovery and justification: Historical and Philosophical Perspectives on the Context Distinction*.

<sup>82</sup> Arabatzis, “On the inextricability of the Context of Discovery and the Context of Justification,” 216.

<sup>83</sup> See Boon et al. “Epistemological and educational issues in teaching practice-oriented scientific research: Roles for philosophers of science,” footnote 3, for an overview of the body of literature in the practice-oriented philosophy of science (Ankeny et al., “Introduction: Philosophy of Science in Practice”) that contributes to better understanding epistemological and pragmatic aspects of knowledge construction.

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